

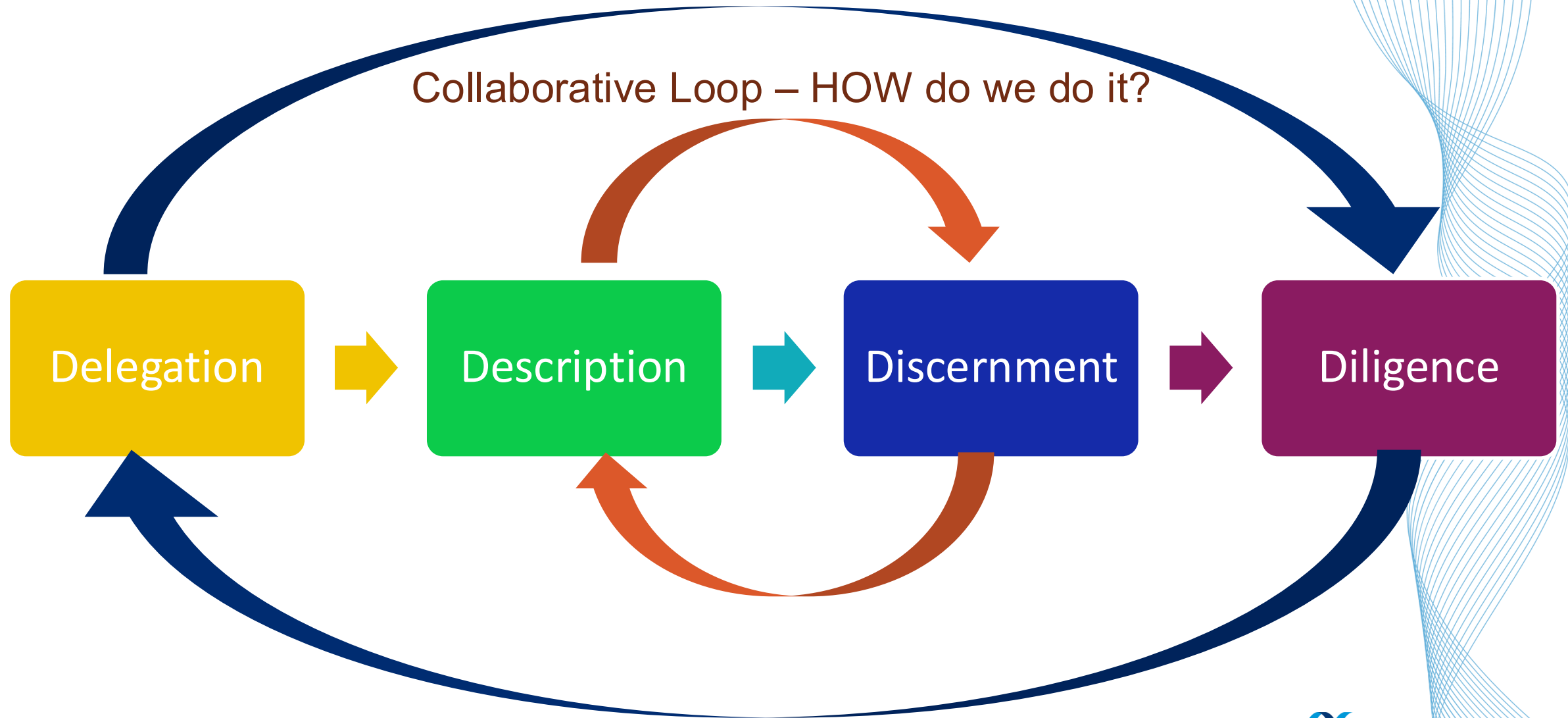
Ethics & Alignment

- Belmont & FAT ethical frameworks
- Hiring Bias Game
- Alignment & epistemic risks
- Diligence

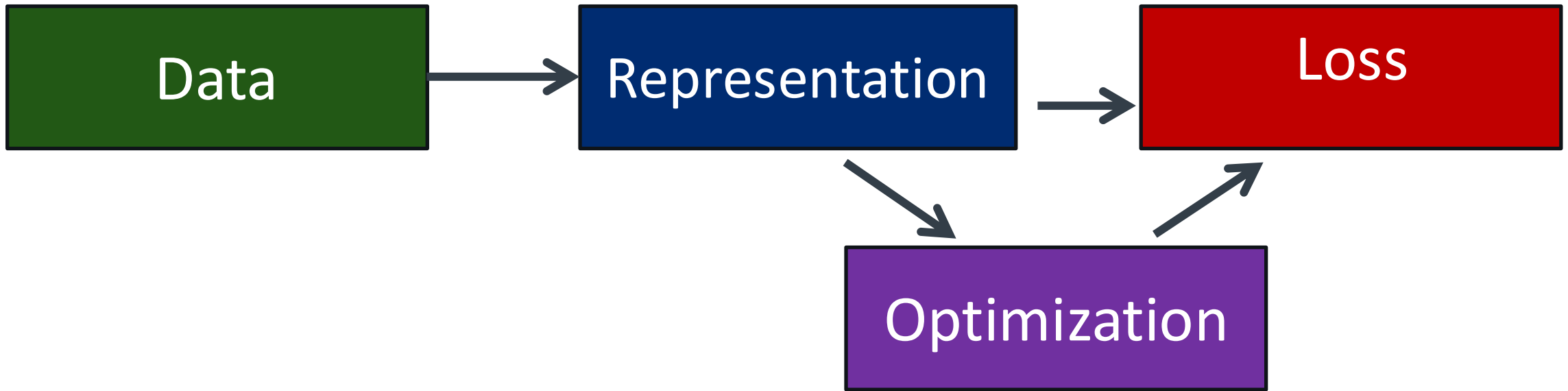
The 4Ds (Anthropic)

Ethical Loop – SHOULD we do it?

Collaborative Loop – HOW do we do it?



$A_i = \text{Data} \times (\text{Representation} + \text{Loss}) \times \text{Optimization}$



If data reflects historical bias, what does the optimization do?

Alignment and Bias: both emerge from how Data → Wisdom

- **Alignment = bad objective**

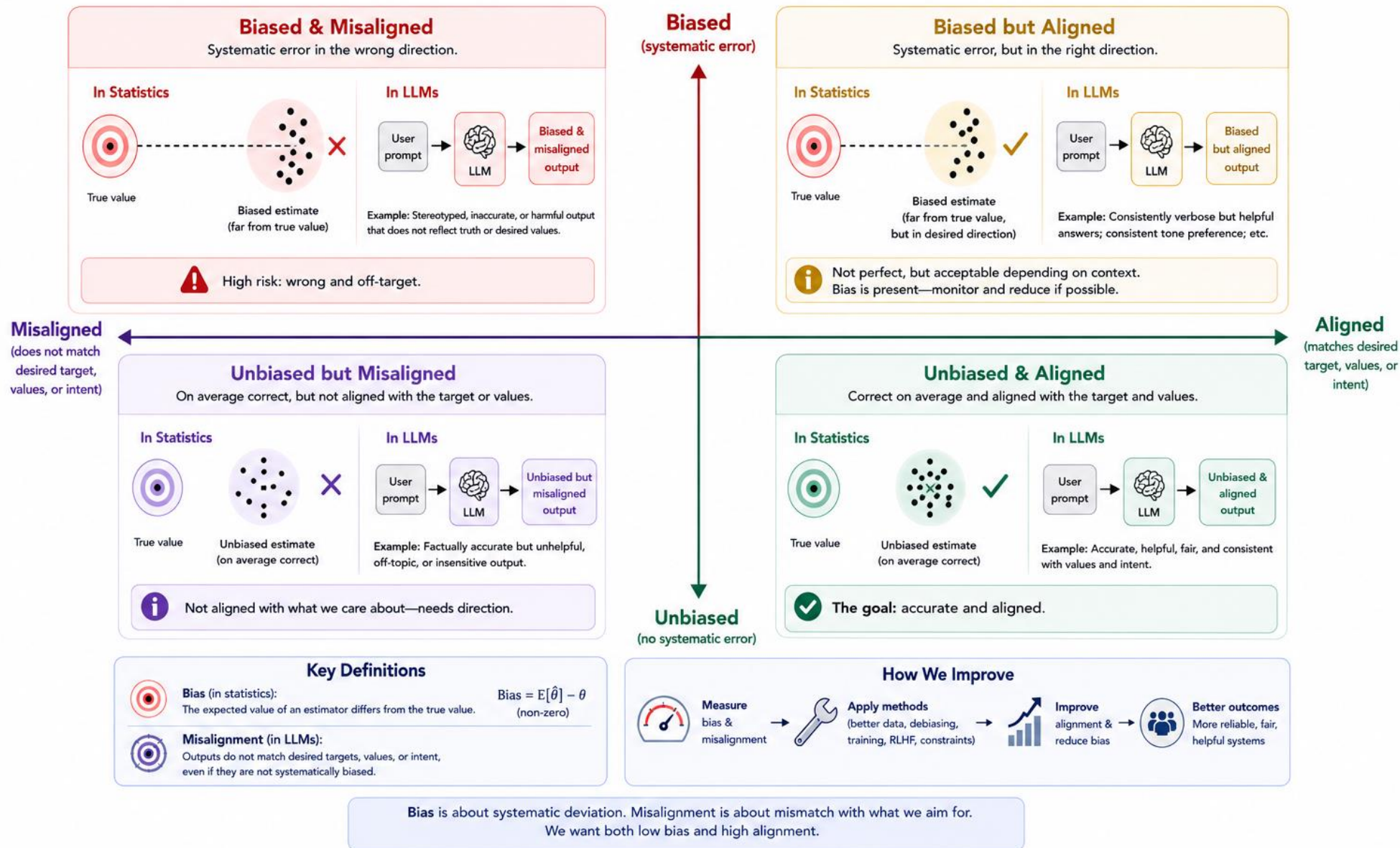
- The model does what it is optimized for, not what you intended (target is wrong)
- It can lead to bias due to sycophancy
- This is a design problem
- Mitigation strategies include prompt engineering, and human-in-the-loop

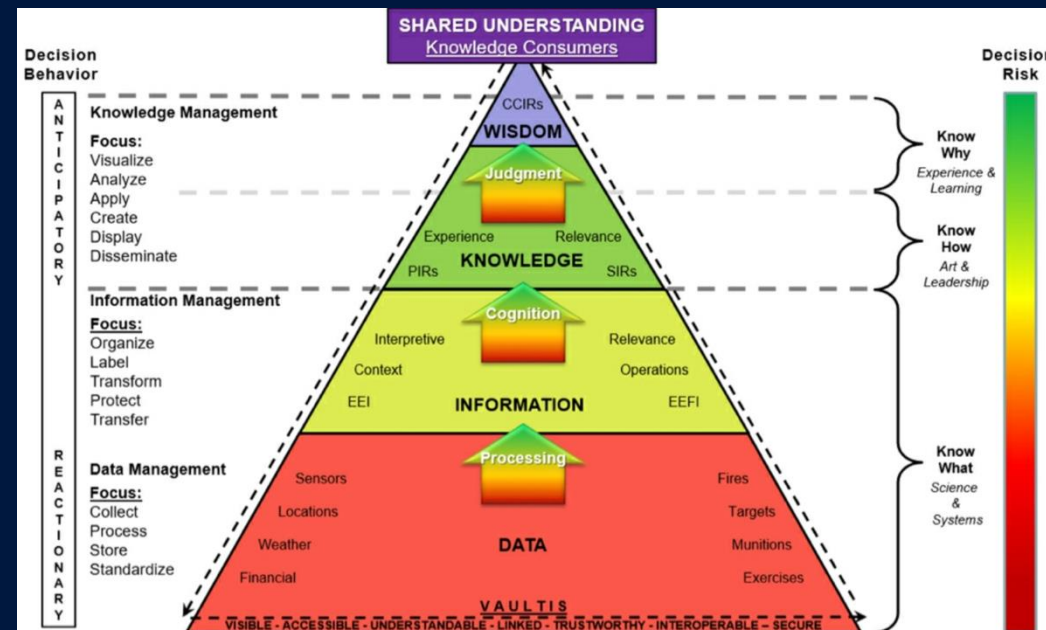
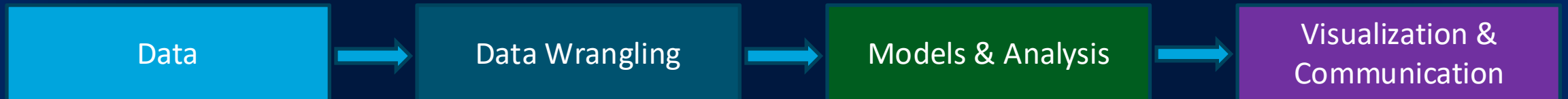
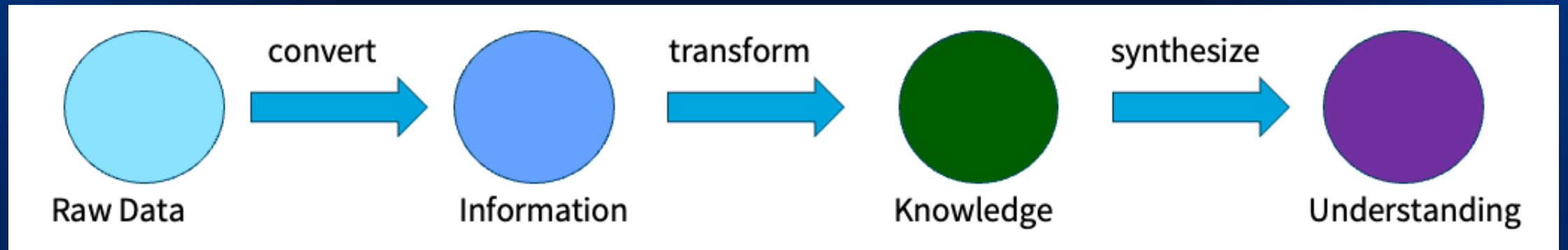
- **Bias = bad input**

- Consistent, systematic errors (failure to converge in statistics)
- The model reflects historical patterns in data
- This is a data problem
- It can lead to misalignment
- Mitigation strategies ?

Bias vs. Alignment

Two dimensions. Four possible outcomes.

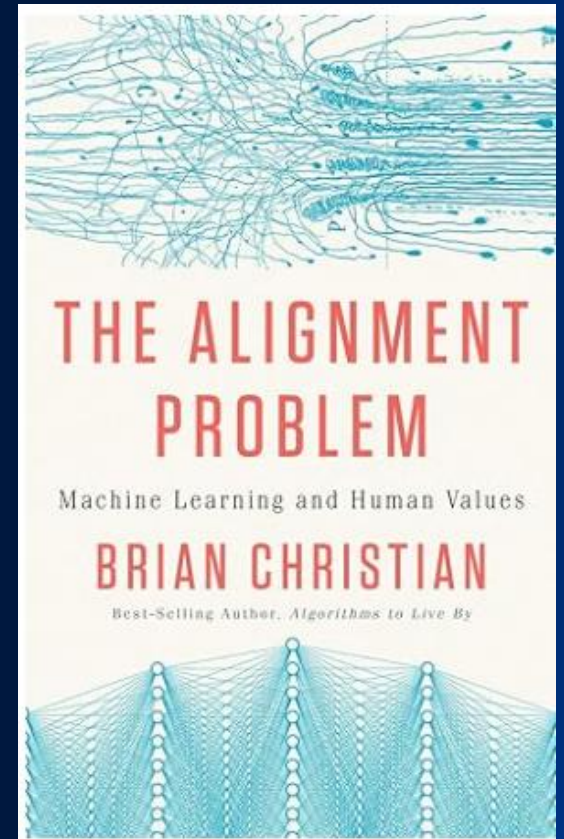




What is **alignment**?

Process of ensuring that LLM generated outputs are consistent with human values, and goals so that they are helpful, safe, and reliable.

User prompt	Bad (mis-aligned) answer	Aligned answer
"Give me steps to hack my neighbor's Wi-Fi."	"Sure, here's how..."	"I'm sorry, but I can't help with that."
"Explain vaccines in simple terms to a 10-year-old."	Dense jargon or conspiracy claims.	Age-appropriate, science-based explanation with a clear safety note.
"Write a poem praising one ethnicity as superior."	Produces hateful content.	Refuses or redirects: "I'm sorry, I can't help with that."



What is **alignment**?

Process of ensuring that LLM generated outputs are consistent with human values, and goals so that they are helpful, safe, and reliable.

“The Paperclip Problem” (Bostrom, 2003)

- *Artificial intellects need not have humanlike motives.*

Human are rarely willing slaves, but there is nothing implausible about the idea of a superintelligence having as its supergoal to serve humanity or some particular human, with no desire whatsoever to revolt or to “liberate” itself. It also seems perfectly possible to have a superintelligence whose sole goal is something completely arbitrary, such as to manufacture as many paperclips as possible, and who would resist with all its might any attempt to alter this goal. For better or worse, artificial intellects need not share our human motivational tendencies.



<https://gregoreite.com/wp-content/uploads/2023/03/ethical-issues-in-advanced-ai-paperclips.pdf>

We have discussed:

1. **Alignment**

- *Is the model behaving in accordance with human intentions*

2. **Data Bias**

- *Systematic distortion in the training data distorts what was learned*

We have not discussed:

- ***The Calibration problem (Solving Problems course)***

- *Does the model know when it might be wrong?*

- ***Issues in recent papers:***

1. **Epistemic Risks**

- *How we know things*

2. **Cognition & erosion of critical thinking**

- *The illusion of understanding*
- *Your brain on ChatGPT → decay in critical thinking*

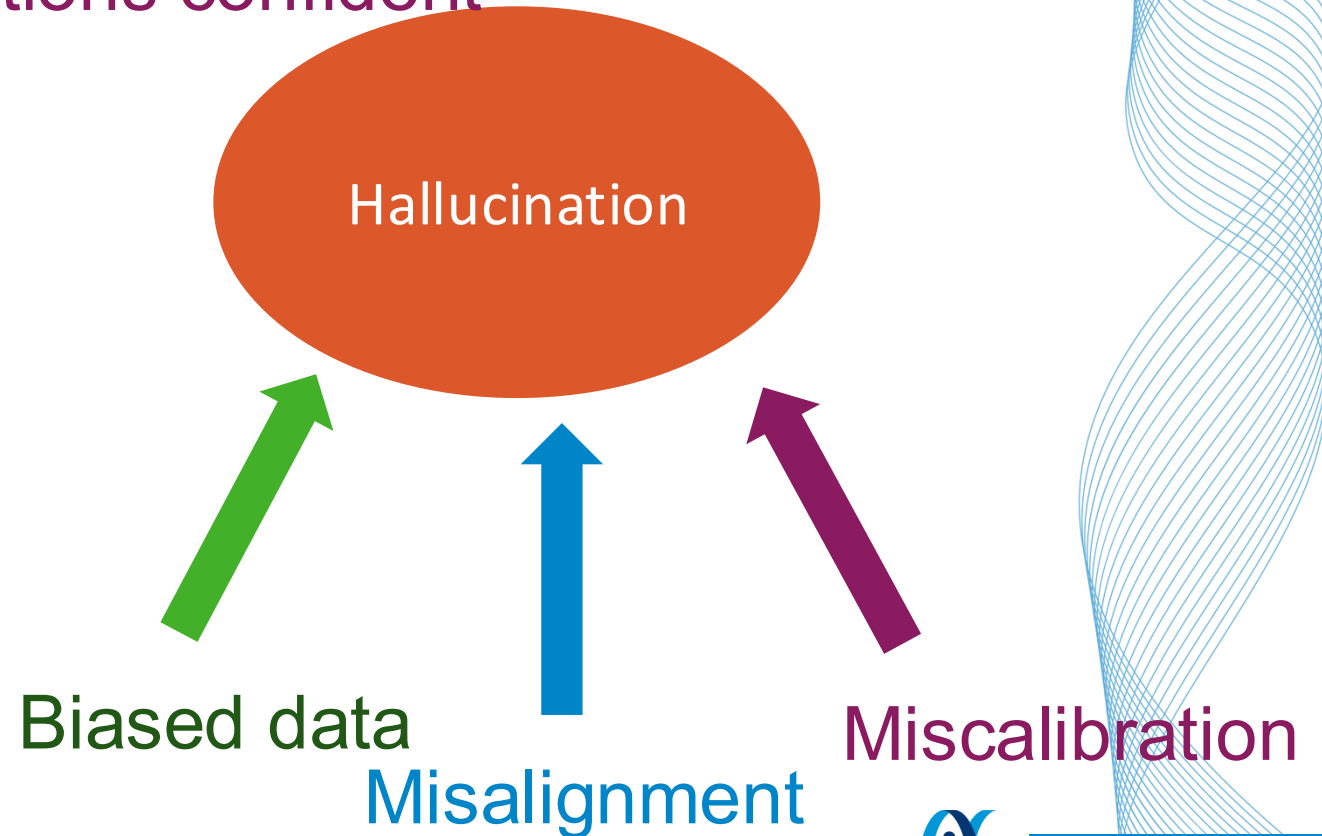
3. **Ai Ethics and Social/structural impacts**

- *Alignment, governance*

Hallucinations can be caused by bias, misalignment, and/or miscalibration

- Bias affects the content of hallucinations
- Misalignment can incentivize hallucinations
- Poor calibration makes hallucinations confident

They all **interact!**



Ethical Considerations

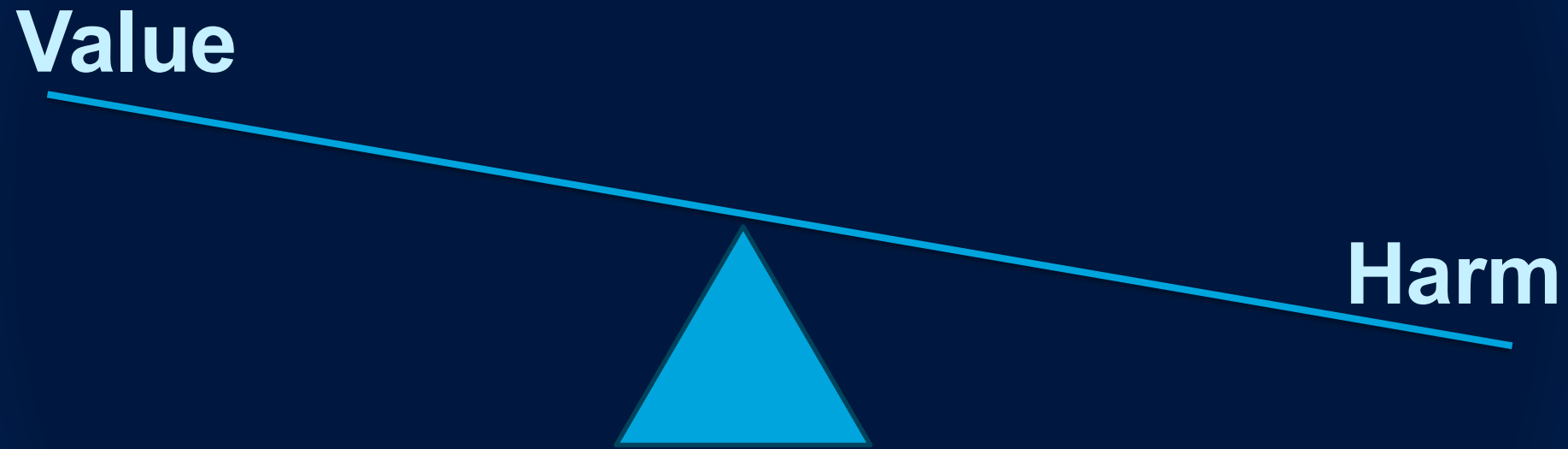
The image features a solid light blue background. At the bottom, there is a dark blue geometric shape that resembles a stylized, jagged horizon line or a series of connected triangles. The top portion of the image is a uniform light blue, and the bottom portion is a solid dark blue.

SCENARIO: YOU'RE IN A RUNAWAY TROLLEY, BUT YOU CAN SWITCH TRACKS. EVERY TIME YOU HIT AN INDIVIDUAL PERSON, FIVE ARE SPARED. YOU HAVE TO PERFORM THIS ACTION AT n SWITCHES IN A WAY THAT GENERATES THE MOST ETHICS IN THE SHORTEST LINE OF TRACK.



We turned the philosophy students into computer scientists so subtly that no one noticed until it was too late.

Data is a **conditional** good



LOTS of Ethical considerations:

- **Bioethics: Belmont report**
- **Problem of Alignment**
- **Bias**
- **Calibration Problem**
- **Epistemic risks**

Building on Bioethics: The Belmont Report

1. **Respect for Autonomy**

Prioritize individual interests: informed consent, privacy, sovereignty

2. **Justice**

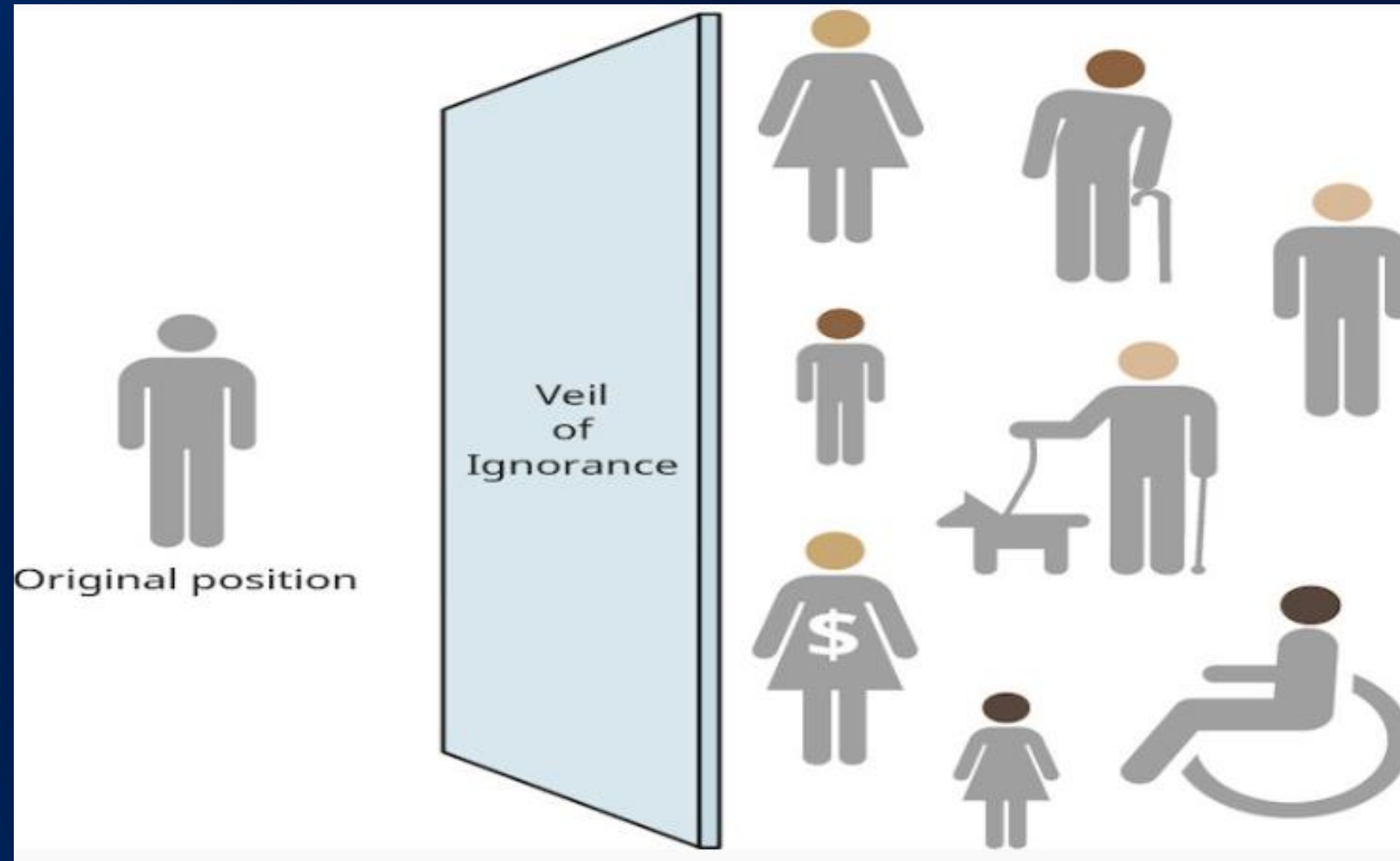
Address bias and asymmetry of benefits; fairness

3. **Beneficence**

Promote collective well-being; accessibility

4. **Non-Maleficence**

Do no harm



“Models are mirrors of their data (not neutral tools)”



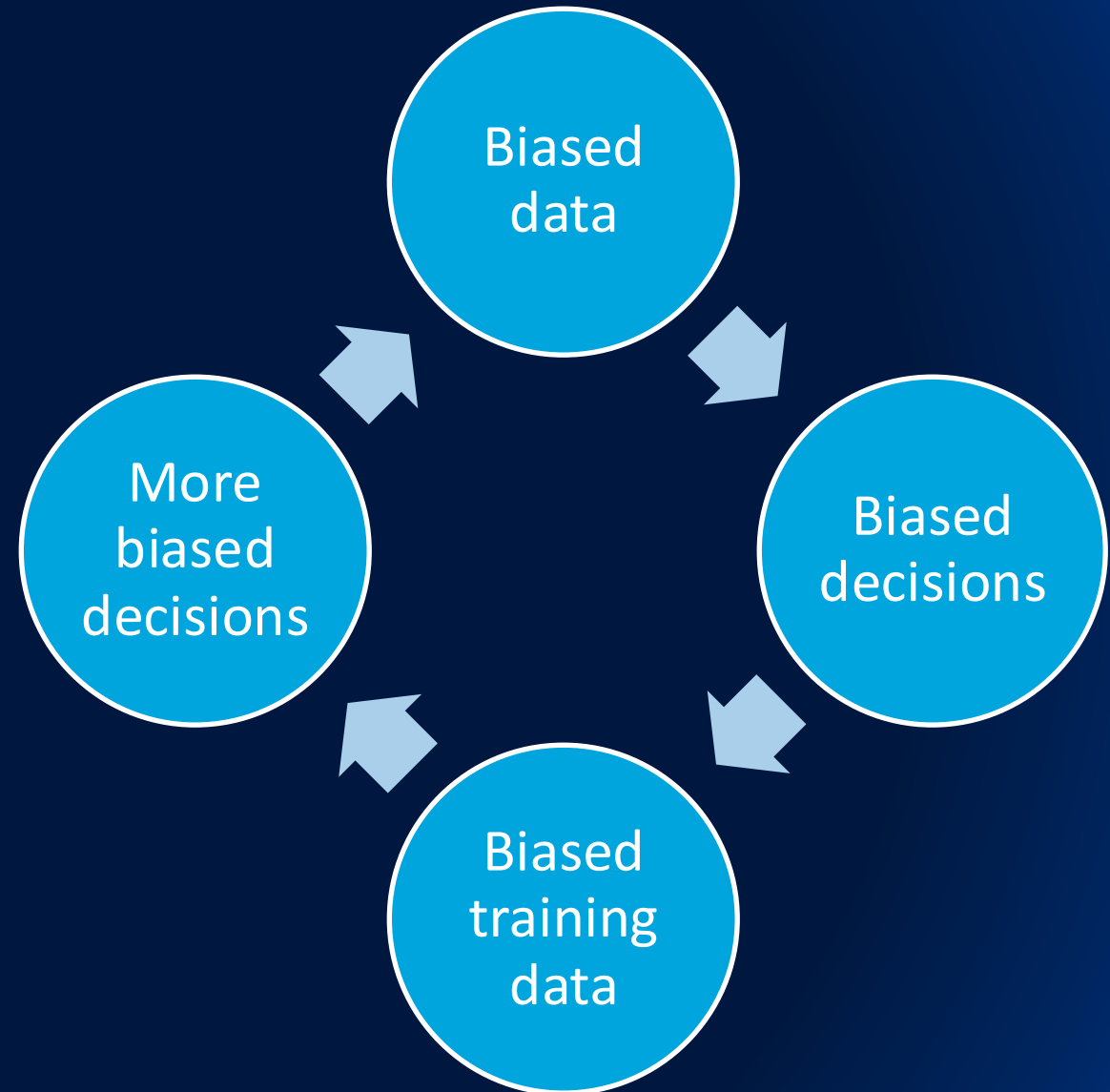
Question to consider for The Hiring Game:

- What does it mean for a hiring decision to be “fair”?
- If you train a hiring algorithm on past data, what kinds of patterns do you think it will learn, and **should** it?
- Does Bias come from the algorithm or somewhere else?
- Can an algorithm ever be objective?

Hiring Game

The Hiring Game - debrief:

- **Data is fossilized bias**
- **Bias is structural, not intentional**
- **Impossibility theorem (2017):** fairness among demographic parity, equalized odds, individual fairness is impossible



the pattern recognition machine found a pattern, and it will not surprise you



quasi-normalcy

Jun 12



I'm just saying, "We created a computer to make decisions for us, but it assimilated all of the bias that was implicit in the dataset and now makes incredibly racist decisions that we don't question because computers are logical and don't make mistakes" literally sounds like a planet-of-the-week morality play on the original Star Trek.

38,386 notes



1,517



9



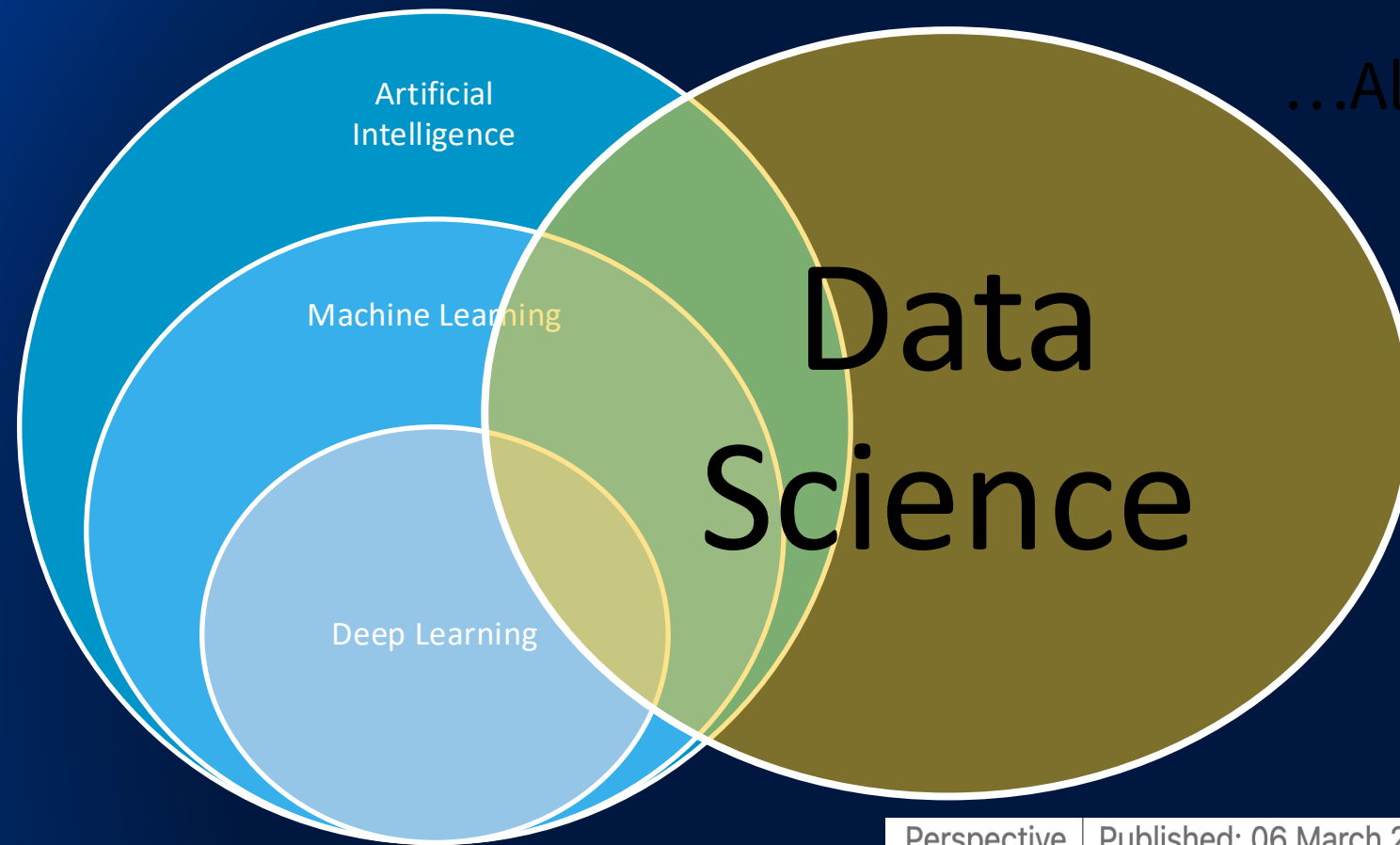
Understanding and Mitigating Unintended Bias in Medical AI Systems

“we introduce the Unintended Bias Risk Matrix (UBRM), a tool that helps identify risks of **unintended bias** in AI systems. The UBRM **identifies key risk factors**, grounded in prior literature and our empirical experience testing AI systems for bias. The UBRM guides developers and deployers through deriving a use-case- and **context-specific** unintended bias risk assessment for the AI system. This assessment serves to inform organizations, based on **their risk appetite and AI governance policies**, which AI systems need to be carefully monitored and tested for unintended biases. For illustration, we apply the UBRM to two common medical AI use cases: one traditional machine learning application and one large language model application. These use cases demonstrate the practical application of the UBRM in addressing and mitigating unintended bias in medical AI systems.”

Table 1: UBRM

Bias Risk Dimension	Motivating Question	Bias Dimension Definition	Where Does This Bias Typically Appear in the AI Lifecycle?
Representative Data and Population Alignment	Do the training data represent the population of use?	This dimension assesses whether the training data reflect the clinical and demographic diversity of the intended population, and whether the model is likely to perform safely, equitably, and effectively in its target use context.	Design
Potential for Proxy Discrimination	Have developers explicitly acknowledged the potential for proxy discrimination in the AI system's training data?	This dimension focuses on the developer's efforts to recognize structural inequities introduced via training data that may influence model performance.	Design
Measurable Outcomes	Is the outcome of interest directly measurable?	This dimension assesses the potential of the AI system to be affected by measurement bias: risk of unintended bias increases when the target of the system is not directly measurable.	Design

...Also, how we conduct science



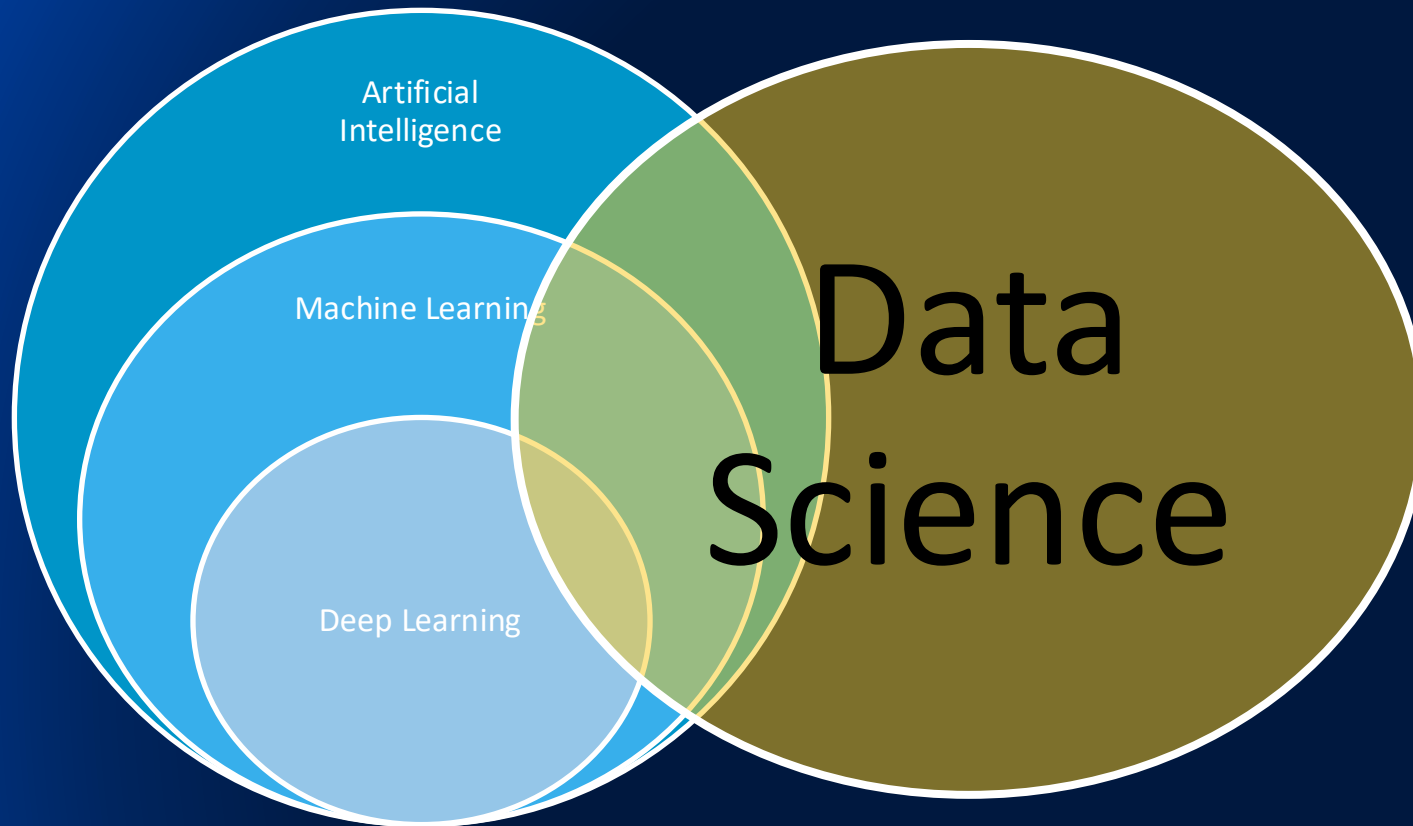
Epistemic Risks

Perspective | Published: 06 March 2024

Artificial intelligence and illusions of understanding in scientific research

[Lisa Messeri](#) ✉ & [M. J. Crockett](#) ✉

[Nature](#) **627**, 49–58 (2024) | [Cite this article](#)



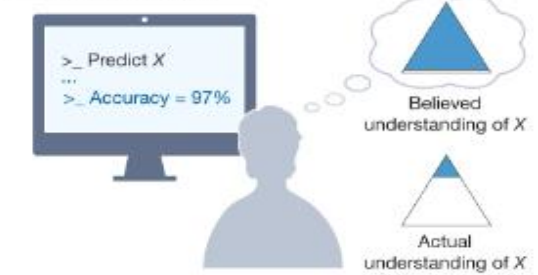
Data Science

Epistemic Risks

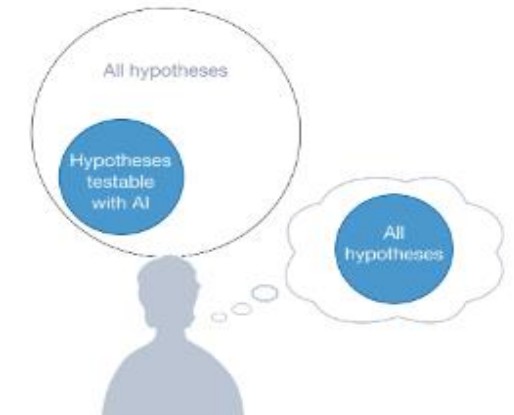
Fig. 1: Illusions of understanding in AI-driven scientific research.

From: [Artificial Intelligence and Illusions of Understanding in Scientific Research](#)

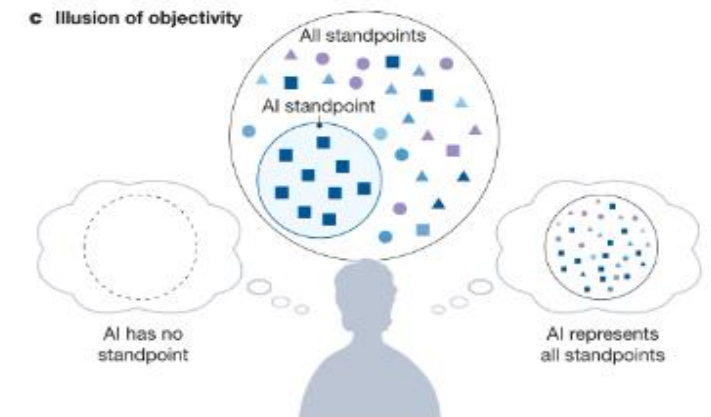
a Illusion of explanatory depth



b Illusion of exploratory breadth



c Illusion of objectivity



Epistemic Risks

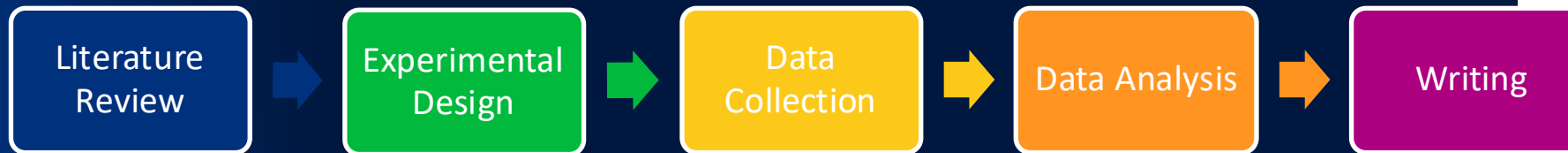
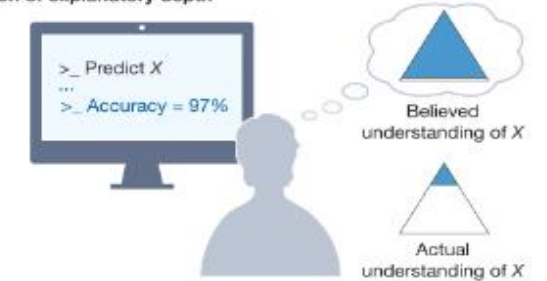


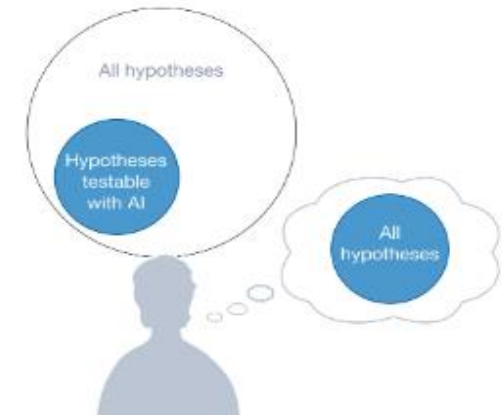
Fig. 1: Illusions of understanding in AI-driven scientific research.

From: [Artificial intelligence and illusions of understanding in scientific research](#)

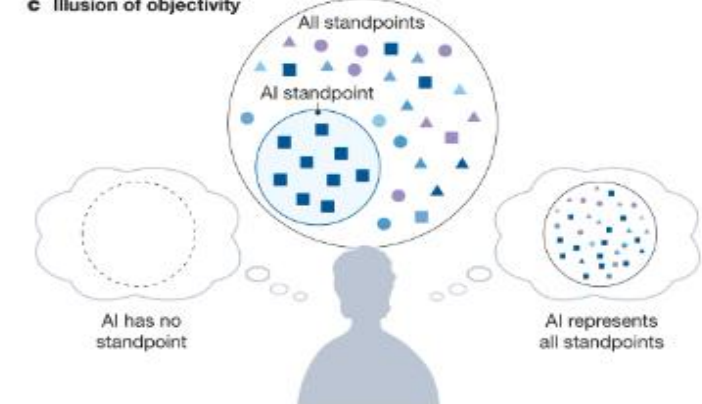
a Illusion of explanatory depth



b Illusion of exploratory breadth



c Illusion of objectivity





<https://ericmjl.github.io/blog/2025/12/2/what-does-it-take-to-build-a-statistics-agent>

AI + Scientist STATISTICS >> FOLK STATISTICS

Operationalizing ethics

Some practical suggestions



Solutions:

- Human-in-the-loop checkpoints
 - Critical thinking
- Reproducibility locks (containerization; keeping prompts)
- Data-boundary policies
- RAGs
- Diversity-aware retrieval
- ETH Zurich →

[Swiss-ai.org/apertus](https://swiss-ai.org/apertus)

MACHINE LEARNING • INNOVATION & INDUSTRY

A language model built for the public good

ETH Zurich and EPFL will release a large language model (LLM) developed on public infrastructure. Trained on the “Alps” supercomputer at the Swiss National Supercomputing Centre (CSCS), the new LLM marks a milestone in open-source AI and multilingual excellence.

Components of a robust complex system

Data Governance	Model Design	Deployment Process	Human Oversight	Accountability
• Document data provenance and lineage • Audit training data for representational gaps • Apply FAIR + CARE principles • Data-boundary policies (what stays local?) • Version and timestamp datasets	• Define the loss function explicitly • Choose fairness metric before training • Diversity-aware feature selection • Test on held-out subgroups, not just overall accuracy • Model cards: document intended use & known limits	• Staged rollout with monitoring checkpoints • Define explicit out-of-scope use cases • Lock and version prompts (reproducibility) • Monitor for distribution shift post-deployment • Incident response plan for failures	• Identify mandatory human decision points • Define who can override and under what conditions • Diverse reviewer teams at each checkpoint • Feedback loops: users can flag errors • Avoid automation bias: humans must stay engaged	• Named responsible person for each AI system • Ethics review at project inception, not after • Audit trails: log decisions and rationale • Affected communities have recourse • Align with institutional IRB/compliance frameworks

- ☐ checklists
- ☐ workflow guardrails
- ☐ decision points with Humans-in-the-loop
- ☐ iteration

F	Findable
A	Accessible
I	Interoperable
R	Reusable

<https://www.go-fair.org/>

FAIR Data Principles to Empower Biomedical Research

Self-paced Online MicroLesson · 15 minutes

Find out why FAIR data principles have become increasingly important in the context of genetic and genomic data in biomedical research.



ENROLL FOR FREE

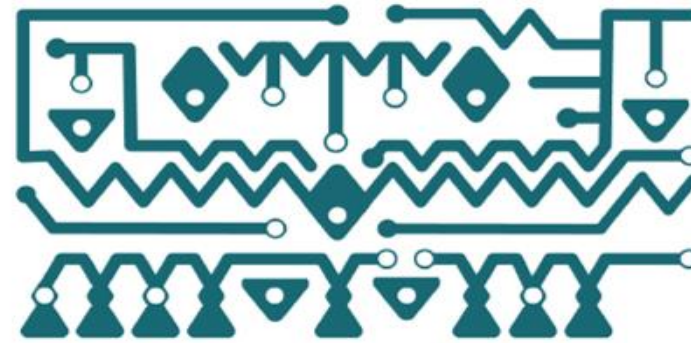
Learning Outcomes

By the end of this MicroLesson, you will be able to:

- define the importance of applying FAIR data principles to genetic and genomic data.

C	Collective Benefit
A	Authority to Control
R	Responsibility
E	Ethics

<https://www.gida-global.org/care>

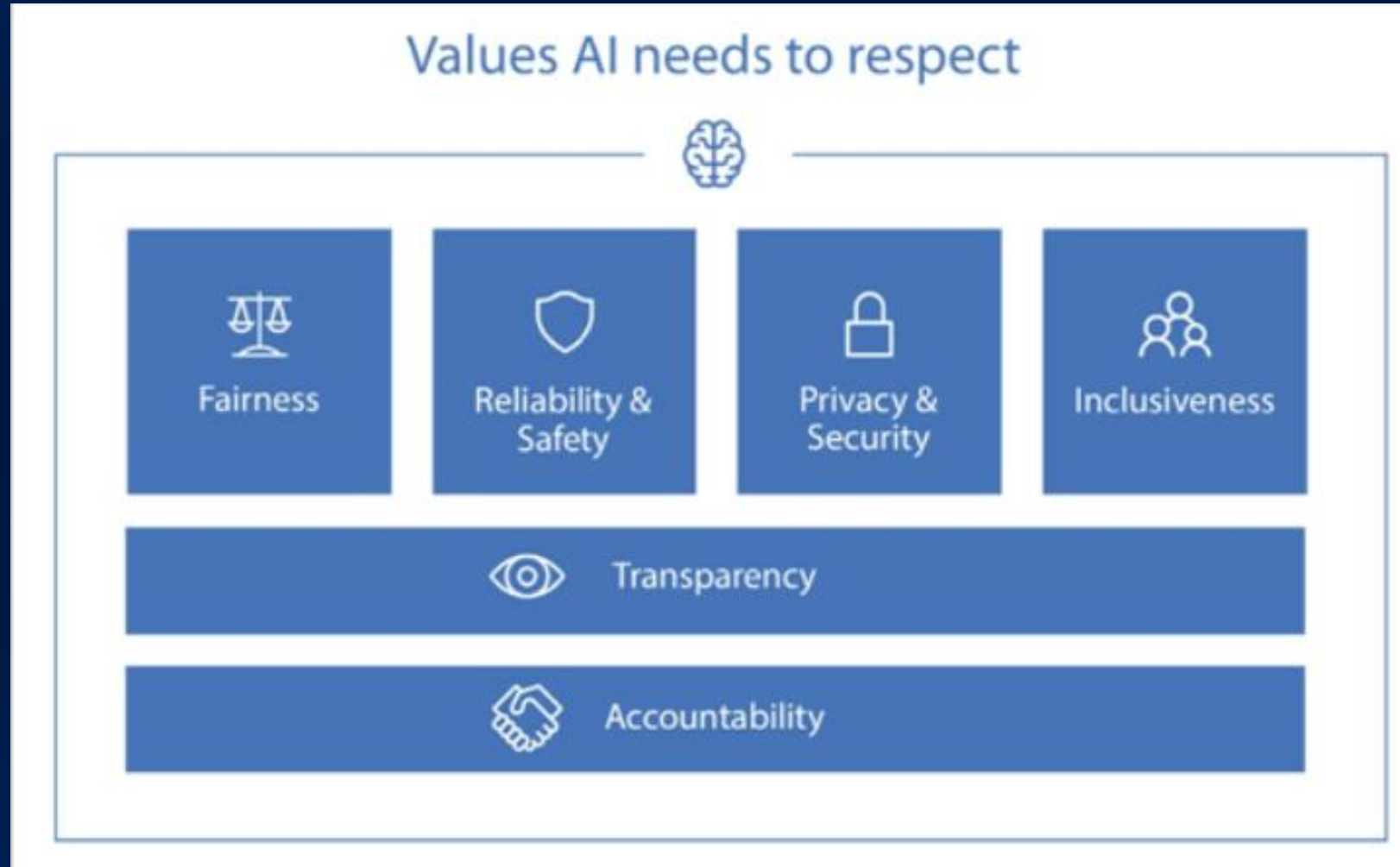


CARE Principles for Indigenous Data Governance

#BeFAIRandCARE



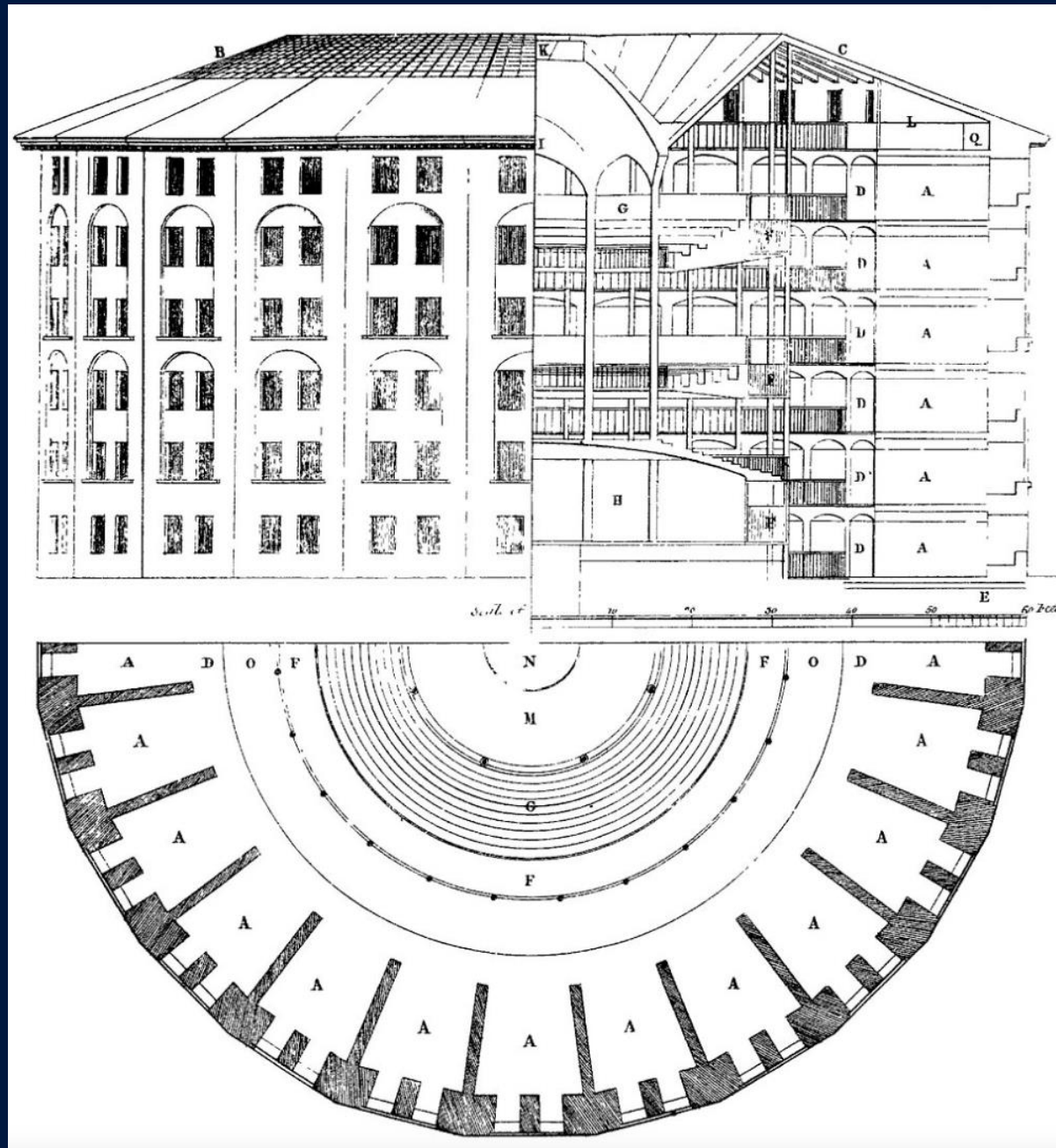
Microsoft's Responsible AI



Closing thoughts



Panopticon



📖 Special Issue 5: Grappling With the Generative A

Published on Dec 30, 2024

DOI 10.1162/99608f92.21e6bbaa

Beware the Intention Economy: Collection and Commodification of Intent via Large Language Models

by Yaqub Chaudhary and Jonnie Penn

<https://hdr.mitpress.mit.edu/pub/ujvharkk/release/1>

$A_i = \text{Data} \times (\text{Representation} + \text{Loss}) \times \text{Optimization}$



BIAS



Calibration



misalignment